

MP461 - Statistical Mechanics

Graham Kells

Module code: MP461

Credits: 5

Semester: 1

Department: ~~THEORETICAL~~ PHYSICS

International: 

Coordinator: Dr Graham Kells (graham.kells@mu.ie)

Overview: Partition functions (simple sub-systems, ideal gas, simple solids, ferro-magnet, Ising model). Micro-canonical and grand canonical ensembles. Phase transitions and critical phenomena. Bose-Einstein and Fermi-Dirac statistics. Bose-Einstein condensation (superconductors and super-fluids). Information theoretic approach to entropy and partition function.

Learning Outcomes

On successful completion of the module, students should be able to:

- Recall the microscopic definitions and basic properties of entropy, temperature and chemical potential
- Give a microscopic interpretation of the second and third laws of thermodynamics
- Deduce and apply the canonical and grand canonical distributions for the states of systems in contact with heat and particle reservoirs
- Calculate partition functions and thermal averages for non-interacting systems such as ideal gases and paramagnets
- Derive the equipartition theorem for the energy and discuss its physical implications and its regime of validity
- Deduce the Fermi-Dirac and Bose-Einstein distributions and apply them in combination with the density of states

Delivery methods	Hours
Lectures	24
Labs / Practicals	0
Tutorials	11
Planned learning activities	0
Independent student activities	90
Total	125

 Assessment

- **Module Graded (numeric value) or Ungraded (Pass/Not Passed):** Graded
- **Continuous Assessment detail(s):** Assignments

Assessment type	Options	Weighting	Duration
Continuous Assessment		20%	
University scheduled written examination		80%	120 minutes
Other		0%	
Total		100%	120 minutes

- **Pass standard:** 40%
- **Could the result of this module be capped?** No

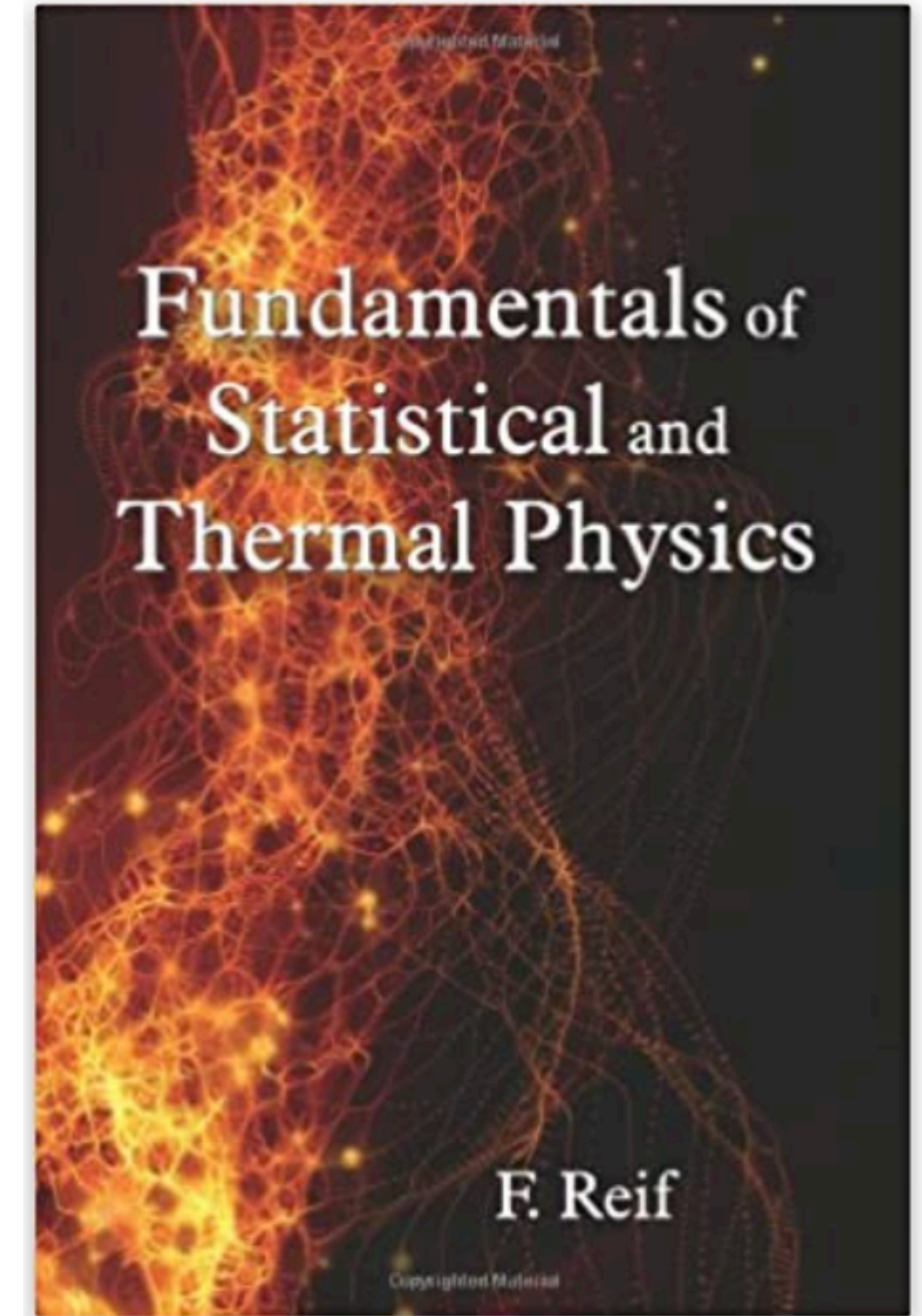
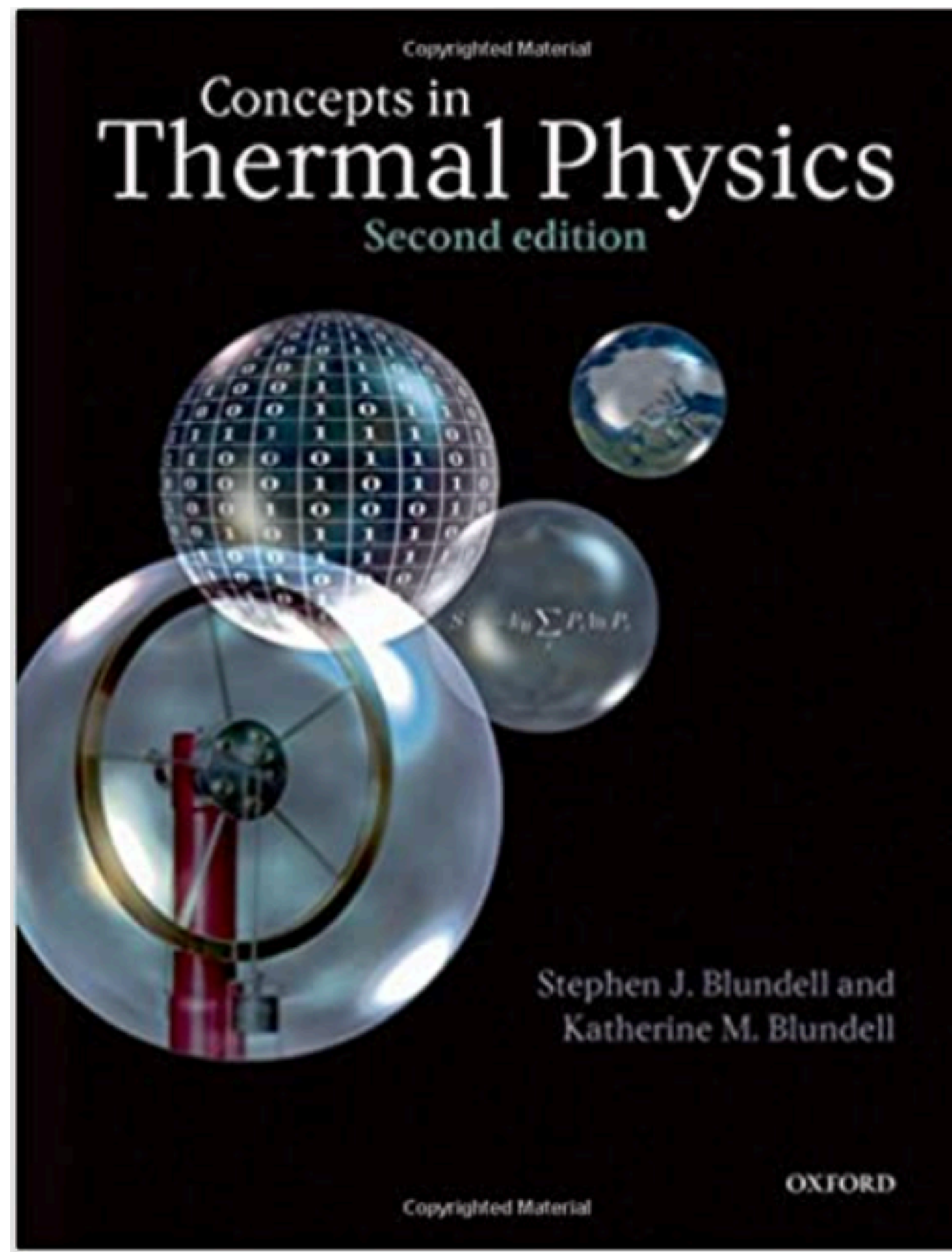
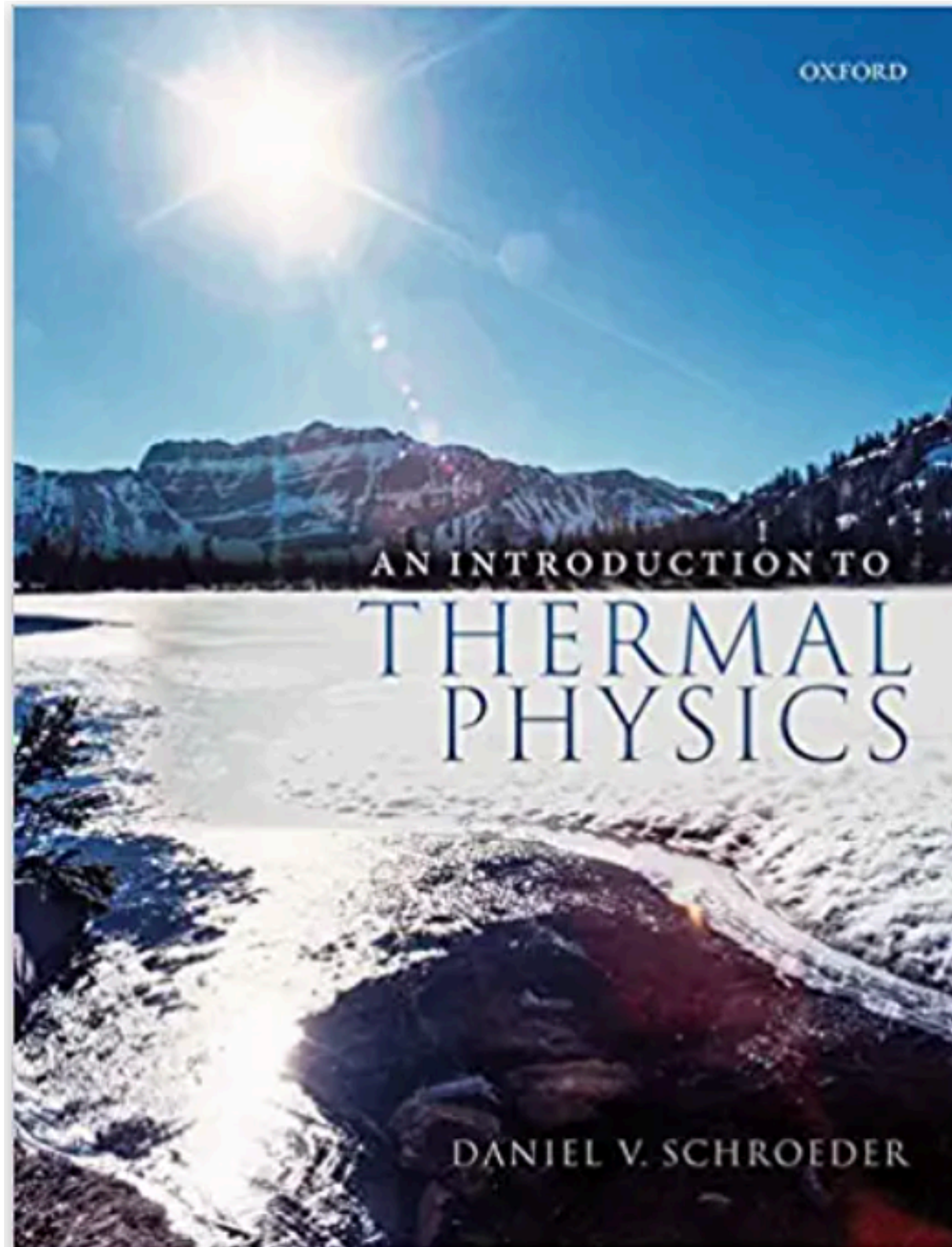
 Autumn Supplementals/Resits

- **Are supplemental/resit registrations permitted?** Yes
- **University scheduled written examination (Autumn):** 120 minutes

 Pre-Requisites

- MP203

Books?



MP203 Refresher

Statistical and Thermal Physics (MP203), *Some notes for review*

Disclaimer: These notes are a learning aid, but do not represent the full content of the course. Formulas and other information besides what is provided here are likely to be useful in passing this course. Any mistakes in these notes are not an excuse to make the same mistake on the exam (check with the book!).

If you have questions about or corrections to these notes, please talk to your lecturer, Graham Kells, (graham.kells-at-mu-dot-ie), Office 1.11 (or 1.9 - not quite clear yet)- Theoretical Physics.

The ideal gas

Schroeder book section 1.2

An ideal gas is an idealized model of a real gas, in which the gas is described as a collection of particles that move freely and do not interact with one another. This is usually a good approximation to real gases at low densities.

The **ideal gas law** is

$$pV = NkT = nRT$$

p is the pressure,

V is the volume,

N is the number of particles of gas,

k is Boltzmann's constant, $k \approx 1.34 \times 10^{-23} J/K$

T is the Kelvin temperature,

n is the number of *mol* of gas

R is the gas constant. $R \approx 8.31 J/K$

The ideal gas law is valid when the gas is in equilibrium.

It is a special case of an *equation of state*. This is a name for any relation between the macroscopic variables of a system which holds in equilibrium states.

For example, a more realistic equation of state for many gases is the van der Waals equation, $(p + n^2a/V^2)(V - nb) = nRT$. This involves the constants a and b , which depend on the actual gas which is being described.

MP203 Statistical and Thermal Physics

Jon-Ivar Skullerud and James Smith

September 17, 2022

Kinetic Theory

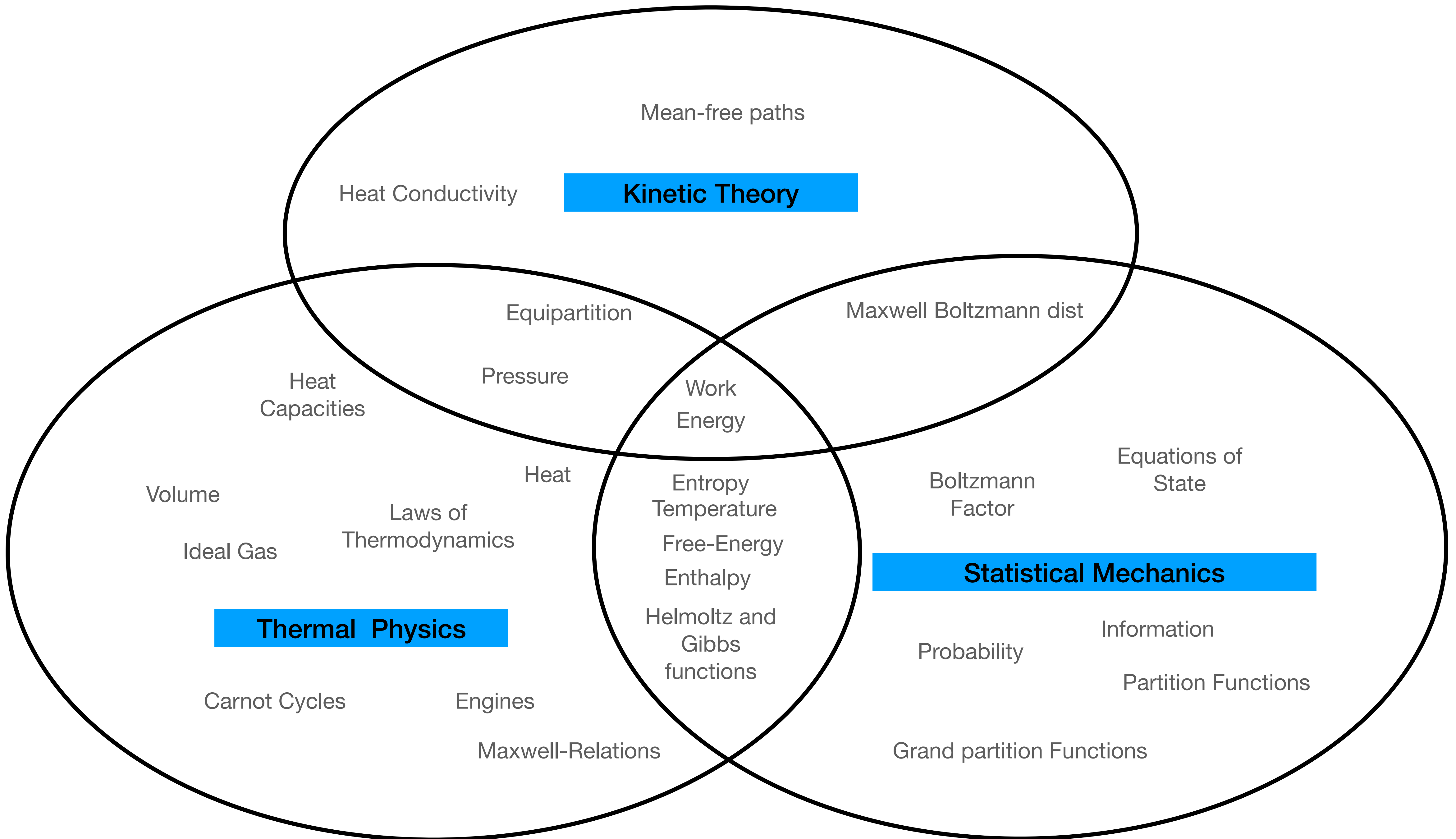
Mean-free paths
Maxwell Boltzmann dist
Equipartition
Heat Conductivity
Ideal Gas
Volume
Entropy
Temperature
Pressure
Laws of Thermodynamics
Carnot Cycles

Thermal Physics

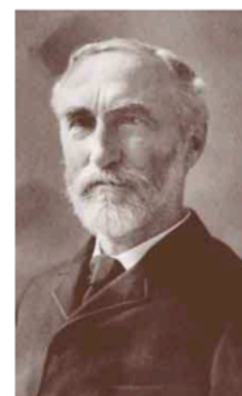
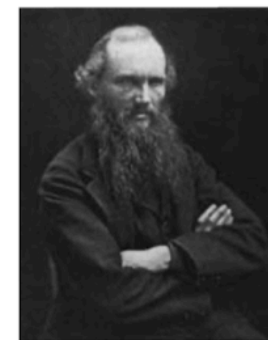
Work Heat
Heat Capacities
Engines
Energy
Adiabats and Isotherms
Equations of State
Enthalpy
Helmoltz and Gibbs functions
Free-Energy

Statistical Mechanics

Boltzmann Factor
Information
Equipartition
Entropy
Probability
Partition Functions
Temperature
Maxwell-Relations



Thermodynamics



Boltzmann

Boyle

Carnot

Clausius

Fourier

Gibbs

Helmholtz

Joule

Kelvin (Thompson)

Lavoisier

Maxwell

Mayer

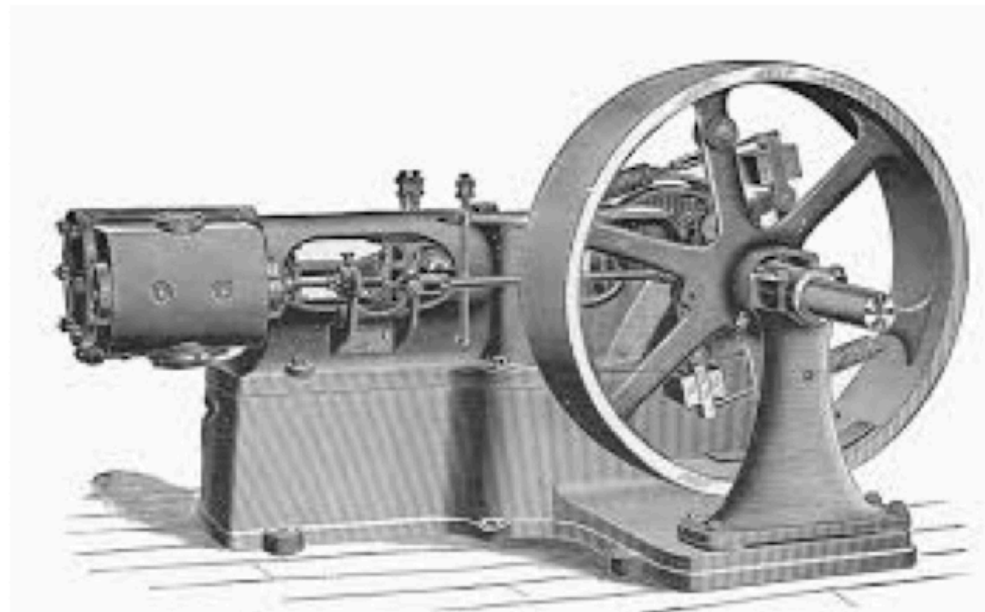
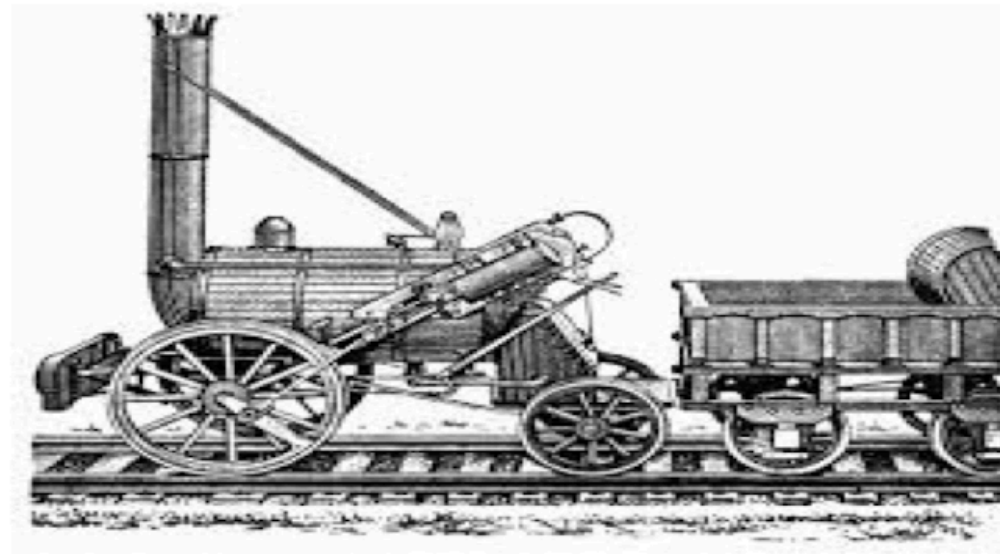
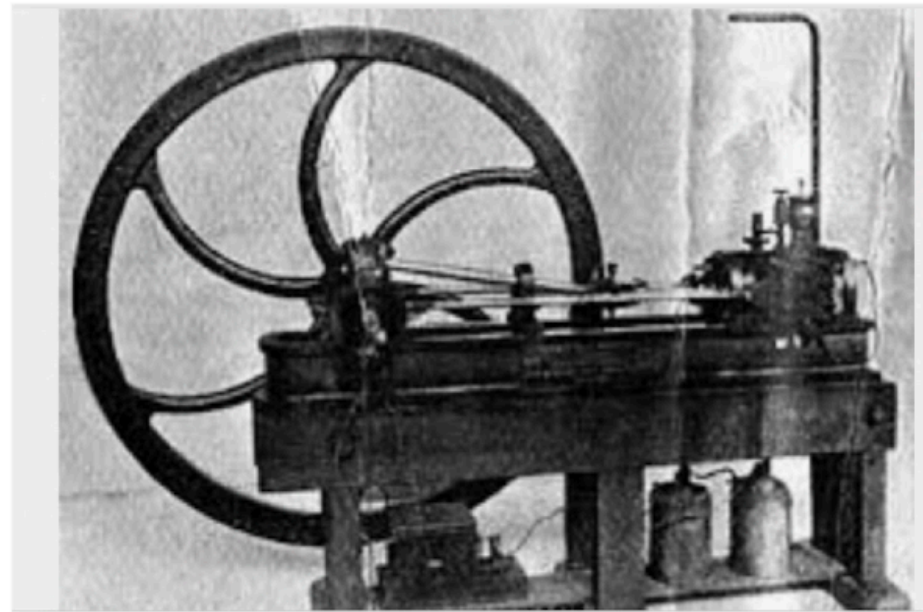
Nernst

Planck

Simon

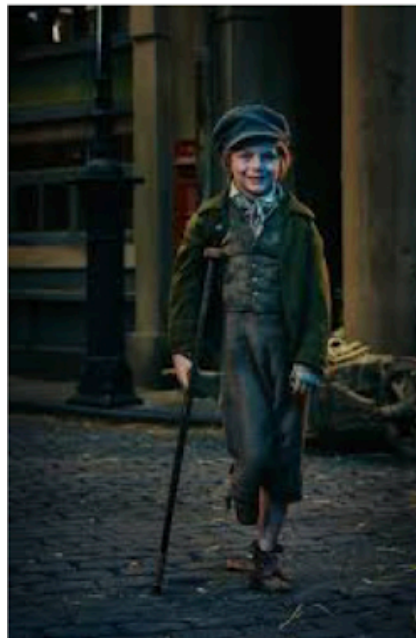
Thompson

Thermodynamics

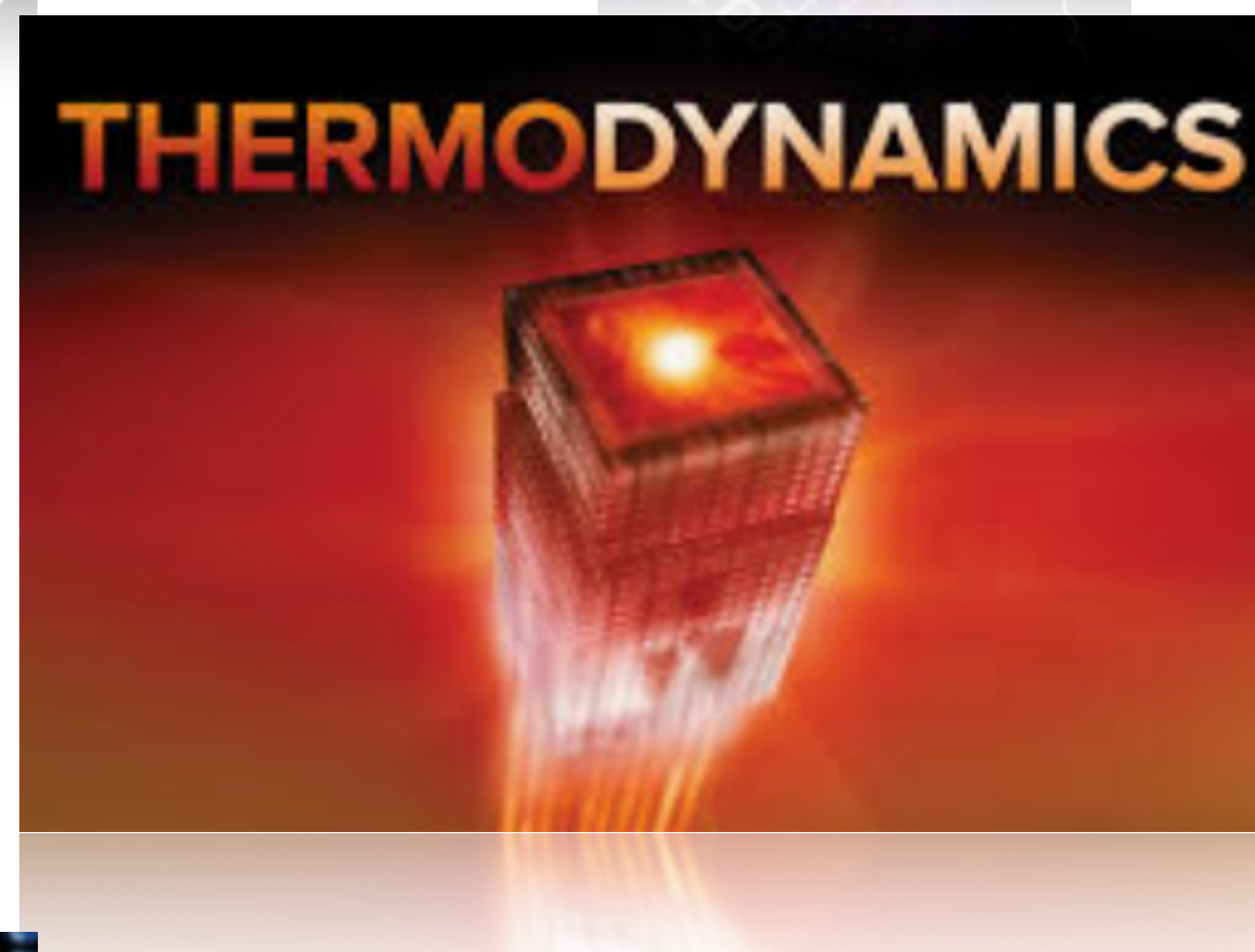


Heat Engines
Carnot Cycles

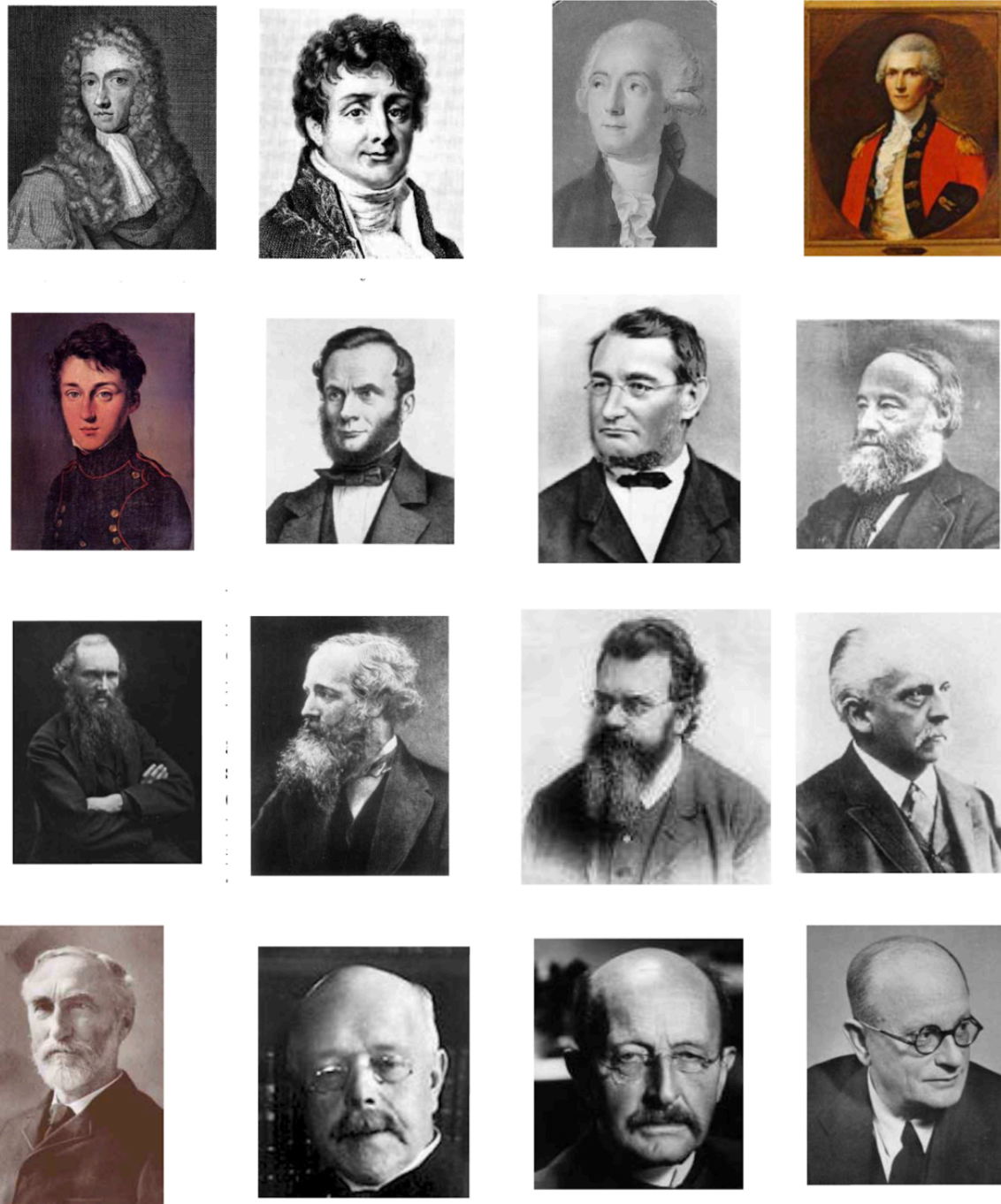
Thermodynamics



Heat Engines
Carnot Cycles



THERMODYNAMICS



0th Law

T Temperature

1st Law

U Energy

2nd Law

S Entropy

3rd Law

Absolute 0

W Work

Q Heat

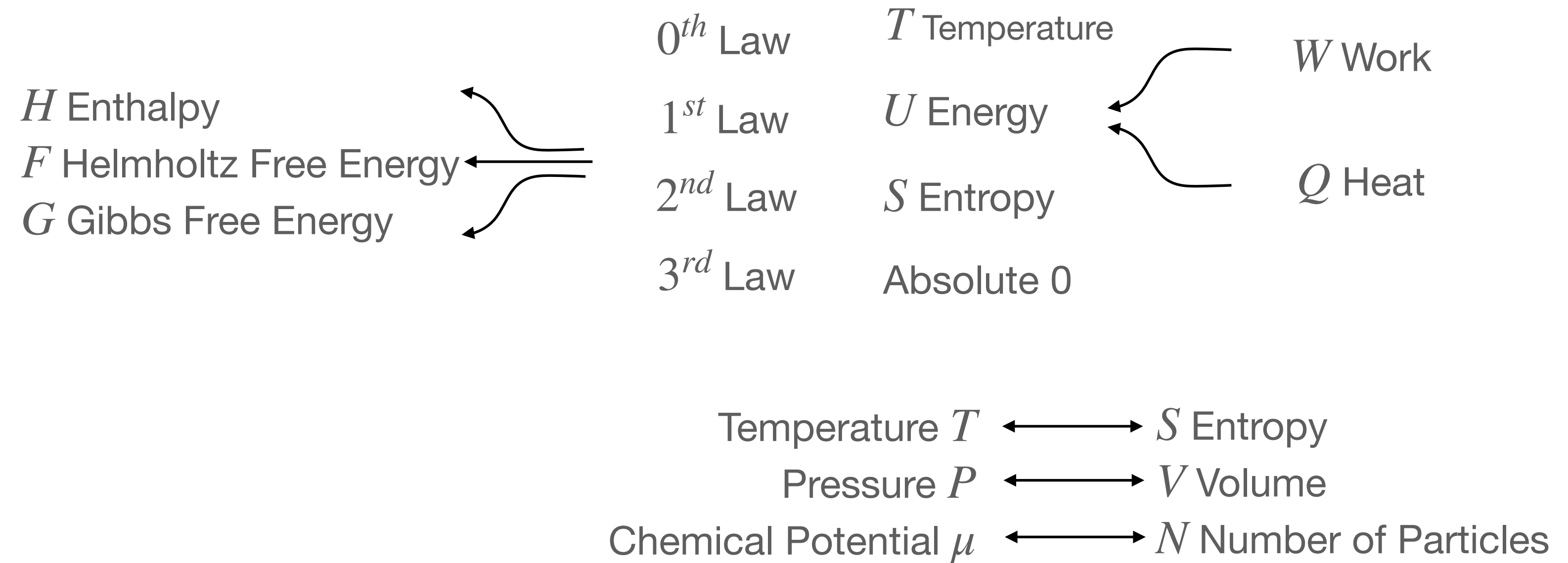
Temperature T $\longleftrightarrow$ S Entropy

Pressure P $\longleftrightarrow$ V Volume

Chemical Potential μ $\longleftrightarrow$ N Number of Particles

$$dU = TdS - pdV + \mu dN$$

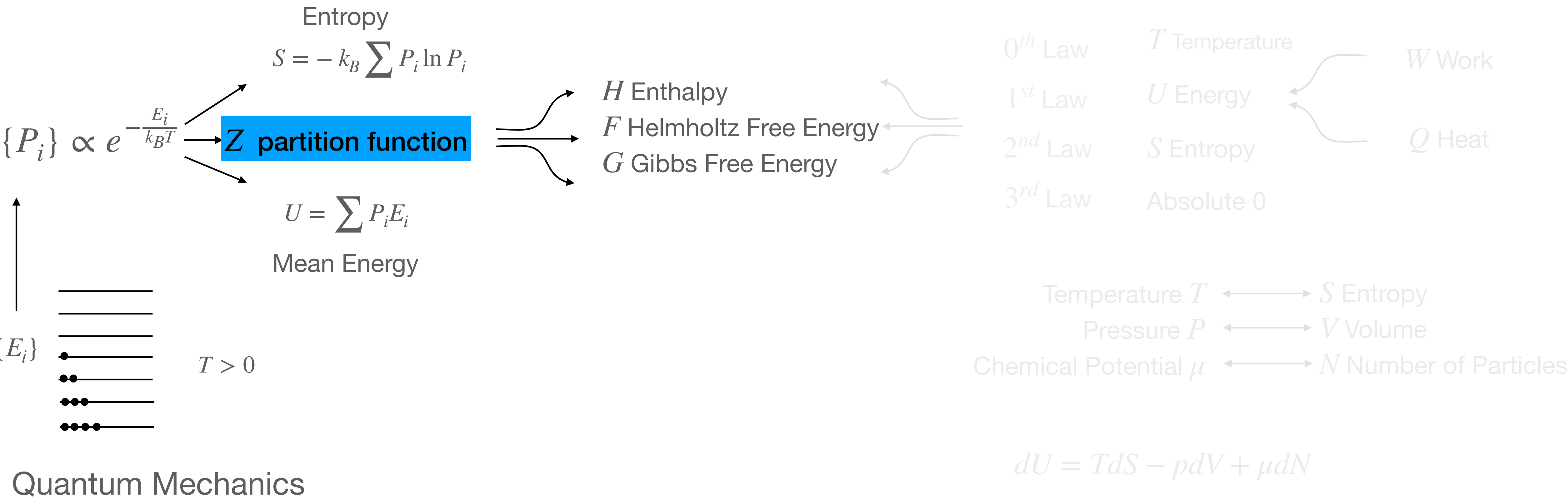
THERMODYNAMICS



$$dU = TdS - pdV + \mu dN$$

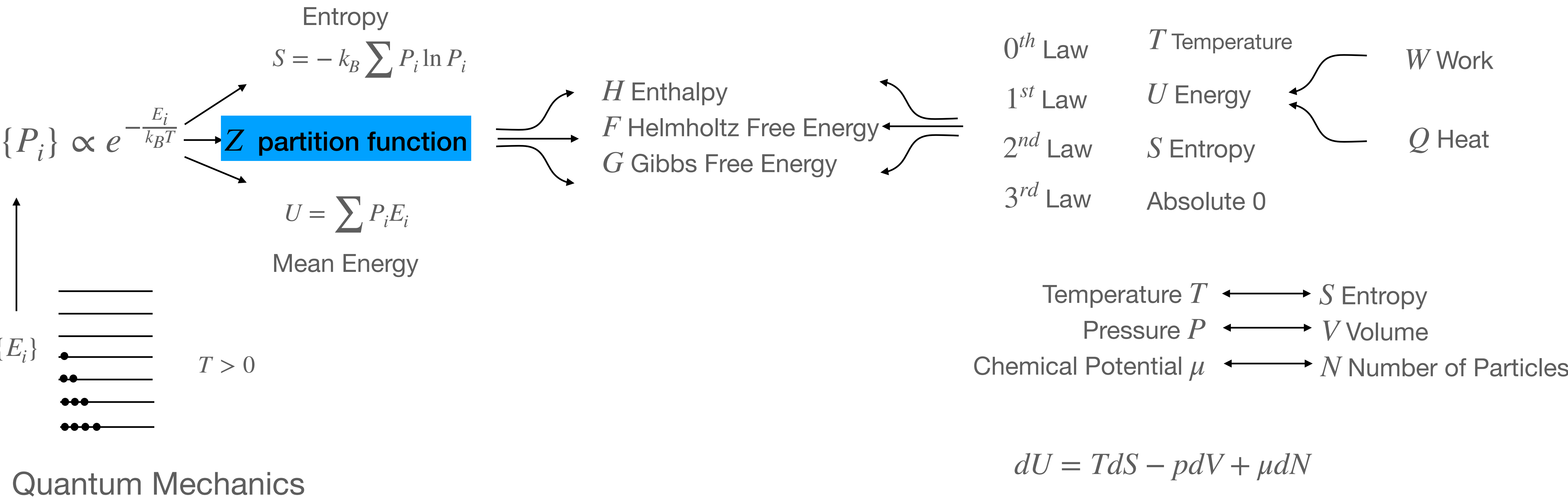
Statistical Mechanics

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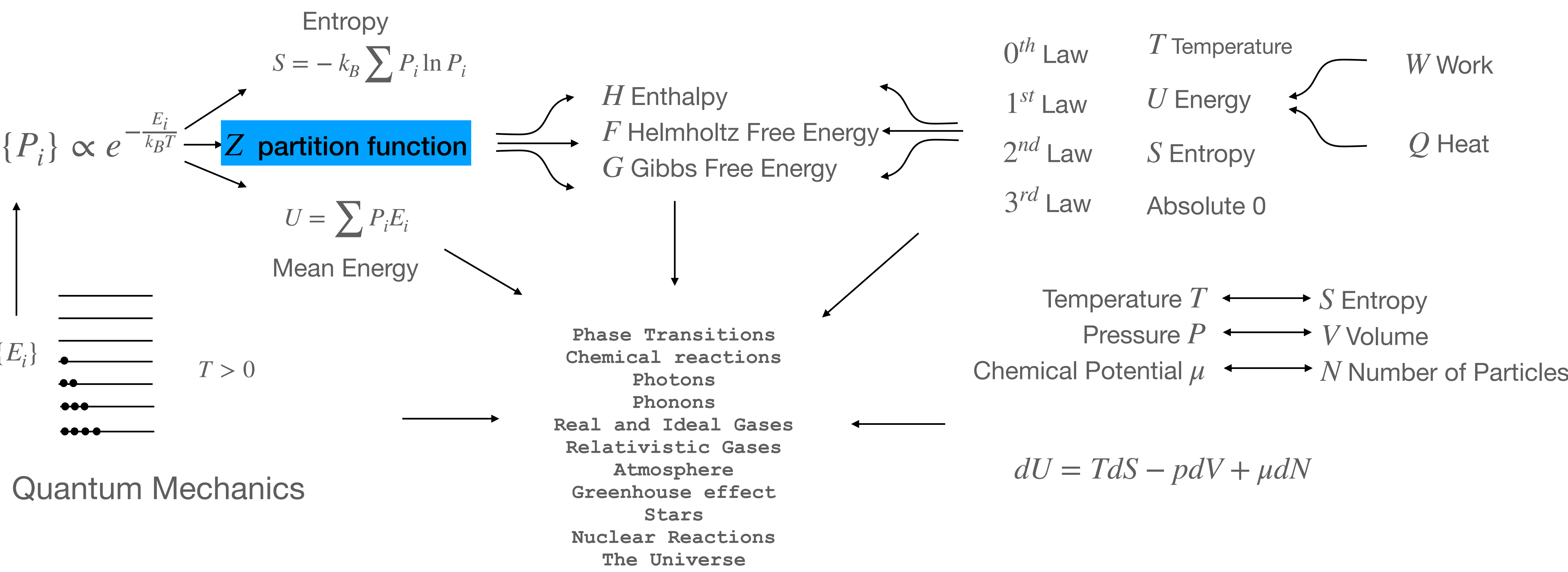
Statistical Mechanics

THERMODYNAMICS



Statistical Mechanics

THERMODYNAMICS



Tentative Schedule

THERMODYNAMICS

Weeks 1-4

H Enthalpy
 F Helmholtz Free Energy
 G Gibbs Free Energy

0th Law

1st Law

2nd Law

3rd Law

T Temperature

U Energy

S Entropy

Absolute 0

W Work

Q Heat

Temperature T ↔ S Entropy

Pressure P ↔ V Volume

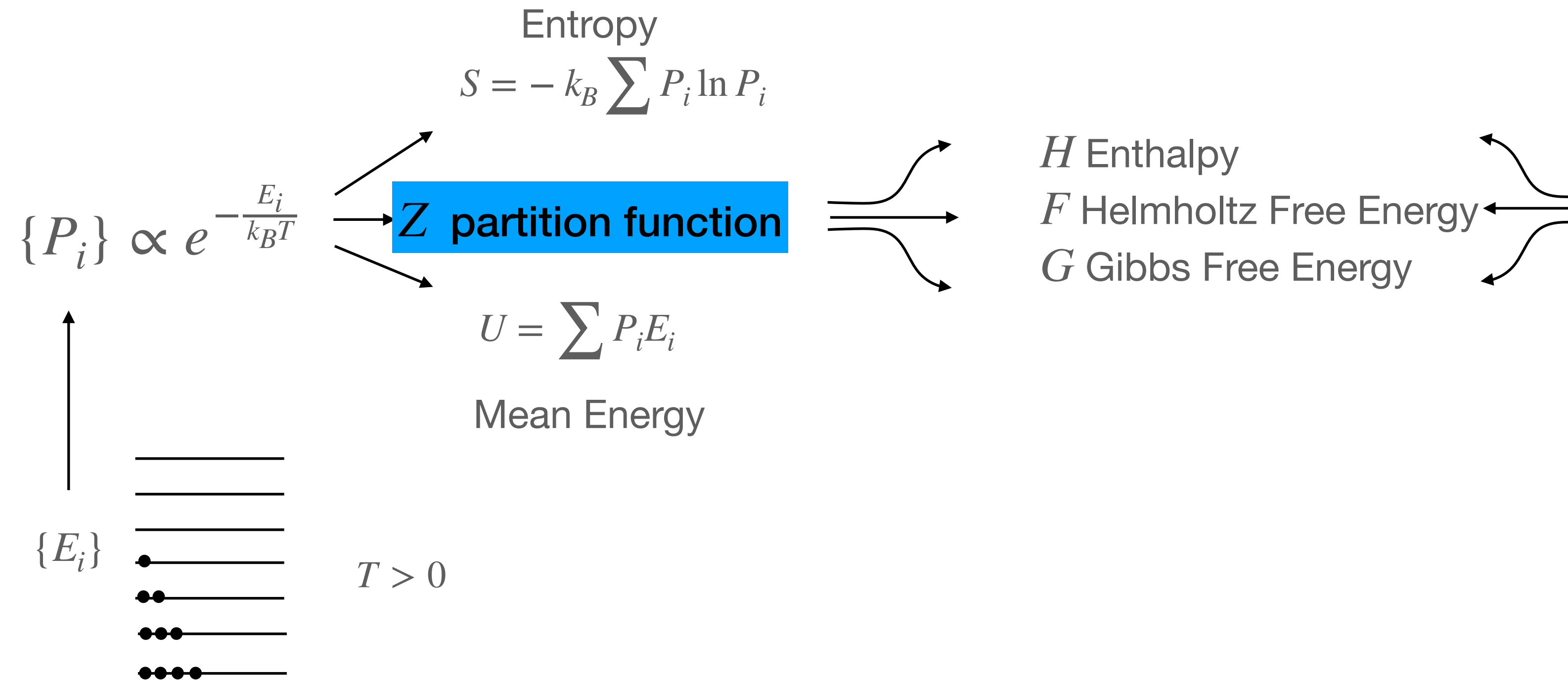
Chemical Potential μ ↔ N Number of Particles

$$dU = TdS - pdV + \mu dN$$

Tentative Schedule

Statistical Mechanics

Weeks 4-9



Quantum Mechanics

Tentative Schedule

Advanced Topics

Weeks 10-12

Chemical reactions
Phase Transitions
Real Gases
Relativistic Gases
Greenhouse effect
Stars
Nuclear Reactions
The Universe